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Acanthopanax senticosus cultures fermented by *Lactobacillus rhamnosus* enhanced immune response through improvement of antioxidant activity and inflammation in crucian carp (*Carassius auratus*)

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A R T I C L E I N F O

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A B S T R A C T

Lactic acid bacteria (LAB) have been recognized as safe microorganism that improve micro-flora disturbances and enhance immune response. A well-know traditional herbal medicine, *Acanthopanax senticosus* (As) was extensively utilized in aquaculture to improve growth performance and disease resistance. Particularly, the septicemia, skin wound and gastroenteritis caused by *Aeromonas hydrophila* threaten the health of aquatic animals and human. However, the effects of probiotic fermented with *A. senticosus* product on the immune regulation and pathogen prevention in fish remain unclear. Here, the aim of the present study was to elucidate whether the *A. senticosus* fermentation by *Lactobacillus rhamnosus* improve immune barrier function. The crucian carp were fed with basal diet supplemented with *L. rhamnosus* fermented *A. senticosus* cultures at 2 %, 4 %, 6 % and 8 % bacterial inoculum for 8 weeks. After trials, the weight gain rate (WGR), specific growth rate (SGR) were significantly increased, especially in LGG-6 group. The results confirmed that the level of the CAT, GSH-PX, SOD, lysozyme, and MDA was enhanced in fish received with probiotic fermented product. Moreover, the *L. rhamnosus* fermented *A. senticosus* cultures could trigger innate and adaptive immunity, including the up-regulation of the C3, C4, and IgM concentration. The results of qRT-PCR revealed that stronger mRNA transcription of IL-1β, IL-10, IFN-γ, TNF-α, and MyD88 genes in the liver, spleen, kidney, intestine and gills tissues of fish treated with probiotic fermented with *A. senticosus* product. After infected with *A. hydrophila*, the survival rate of the LGG-2 (40 %), LGG-4 (50 %), LGG-6 (60 %), LGG-8 (50 %) groups was higher than the control group. Meanwhile, the pathological damage of the liver, spleen, head-kidney, and intestine tissues of probiotic fermentation-fed fish could be alleviated after pathogen infection. Therefore, the present work indicated that *L. rhamnosus* fermented *A. senticosus* could be regard as a potential intestine-target therapy strategy to protecting fish from pathogenic bacteria infection.

1. Introduction

Currently, there has been an increasing amount of literatures on aquatic animals infected with *Aeromonas* spp., especially in freshwater fish, which was regarded as an emerging opportunistic pathogen in aquaculture [1]. The septicemia, gastroenteritis and respiratory infectious disease could be driven by the motile *Aeromonas* species in fish and humans, particularly *Aeromonas hydrophila*, *Aeromonas veronii*, and *Aeromonas sobria* [2]. *A. hydrophila* infection mainly correlated with soft tissue damage and skin wound, and the interaction between host micro-environment and pathogen could be triggered by the extracellular proteins (ECPs) of *A. hydrophila* through alteration of cytokines, chemokines, and mucosal immunoglobulin [3]. Regarding the potential public health concerns, it is urgent to develop novel strategy to prevent transmission and persistence by *A. hydrophila* in the aquaculture.

Recent research developments have revealed that the intestinal

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physiology functions and immunological homeostasis could be maintained by gut microorganisms, which play a vital role in regulating the stability of local intestinal microecosystem [4,5]. Nevertheless, the external environmental factors and pathogenic bacteria often result in microecological imbalance and immune-related inflammatory disorder, which could be attributed to the changes of commensal microbial community structure and function [6]. A disturbance in the intestinal flora could lead to the penetration of the physical and immunologic barrier by opportunistic pathogens, which would take advantage of these situations to reduce the integrity of the epithelium and cause severe inflammation [7,8]. Therefore, regulating the homeostasis of intestinal commensal microbiota and mucosal barrier function after bacterial infection requires further consideration [9]. *Aeromonas* spp. Infected fish specifically susceptible to hemorrhagic and diarrheal diseases, is a notable model for investigating mucosal microecology and immune homeostasis in vertebrates [10]. To prevent *Aeromonas* species infection and reduce the invasion of enteric pathogenic or symbiotic bacteria, antibiotics were widely applied in feeds of aquatic animals for prevention and treatment of *A. hydrophila* [11].

However, the investigators have understood that antibiotic abuse can damage normal flora structure in the gastrointestinal tract, in addition to killing of pathogen. Excessive use of antibiotics, particularly in the early life stages of aquatic animals, may be lead to drug-resistant bacteria, water environment contamination, and food safety concerns [12,13]. In addition, fish frequently uptake drugs supplied feed result in water environment as a major reservoir for antibiotic, which aggravates the more widespread diffusion of medicine molecules into various aquatic animals [14]. As important composition of commensal microbiota, several probiotics have been demonstrated replacement with antibiotic therapy in terms of promoting intestinal growth, preserving immunological homeostasis, and regulating the intestinal microecosystem [15,16]. Notably, modulation of commensal flora by probiotics could enhance intestinal immunity through a variety of mechanisms, such as mitogen-activated protein kinase (MAPK), and nuclear factor kappa-B (NF-κB) pathways [17].

Lactic acid bacteria (LAB), a type of anaerobic bacterium, are recognized as safe microorganisms that improve micro-flora disturbances, which widely inhabit in intestine of fish. Recent evidence has suggested that LAB may be an enhancer of the gastrointestinal and immune system, which can facilitate antigen delivery on the mucosal surfaces, and antagonistic bacterial interactions rather than immune tolerance [18]. While antibacterial compounds release from LAB, and enzymatic activities of probiotics in the intestine, are easily affected by the microbial colonization and gut physiology. Typically, *Lactobacillus rhamnosus* could competitively inhibit pathogens and strongly adhere to intestinal epithelial surfaces and mucus, which depends on micron-long pili expressing fimbriae, adhesins and outer membrane proteins [19,20]. Currently, investigators have demonstrated the effects of traditional Chinese medicine fermented product on inflammatory bowel disease (IBD), ulcerative colitis (UC) and infectious diseases [21,22]. Furthermore, the beneficial effects on cancer, gastroenteritis and metabolic diseases modulated by herbal medicine fermentation with probiotics have been reported, which is attributed to microbe promote the metabolites production and transform glycosides to aglycones [23]. *Acanthopanax senticosus* is a well-know traditional herbal medicine that extensively utilized in aquaculture, which could improve appetite loss, fatigue, growth performance and inflammation [24]. Nonetheless, few studies explore the mechanisms of *A. senticosus* fermentation with probiotic to modulate intestinal barrier function in fish. Moreover, different isolates belonging to the same species may result in distinctive biological effects, which require further research to provide evidence for potential explanations.

To elucidate the effects and mechanisms of *A. senticosus* fermentation with *L. rhamnosus* (LGG) on the growth performance and immune response of crucian carp, the effects of LGG fermented *A. senticosus* cultures on the growth performance, immune response, and disease resistance in fish were investigated. Therefore, the present work would generate fresh insight into stimulation induced by probiotic fermented Chinese herbal cultures, an effective and safe strategy for maintaining intestinal barrier function in vertebrates.

2. Materials and methods

2.1. Fish maintenance and ethics statement

Crucian carp weighing 20.0 ± 3.0 g with a length of 8.0 ± 0.5 cm were supplied by the aquaculture in Changchun, Jilin Province, China. The fish were kept at 27.0 ± 1.0 °C in cylindrical tanks (80 cm diameter) with flow-through raring system. Fifty fish were cultured in each tank with a continuous flow of filtered and oxygenated water. During experiment periods, water pH was kept at 7.8 ± 0.6, nitrate at 0.015 ± 0.002 mg/L, dissolved oxygen at 5.5 ± 0.45 mg/L, and total ammonia at 0.13 ± 0.02 mg/L. Fish were acclimatized to laboratory conditions for one week and fed twice a day with basal diet at 3 %–4 % body weight. All animal experiments were performed following the Jilin Agriculture University Institutional Animal Care and Use Committee (JLAU08201409), and the experimental procedures were performed in accordance with the National Institutes of Health guide for the care and use of Laboratory animals (NIH Publications No.8023).

2.2. Bacterial strains and growth conditions

L. rhamnosus ATCC 53103 (LGG) is purchased from American type culture collection (ATCC), a plasmid-free strain cultured anaerobically in De Man, Rogosa and Sharpe (MRS) medium (Oxoid, UK) at 30 °C, isolated from human feces. Virulent *A. hydrophila* TPS-30 was isolated from diseased common carp (Bioproject accession: PRJNA381370), and grown in LB broth at 30 °C.

2.3. Preparation of fermentation cultures and diets

The LGG were diluted at 10^{-5} CFU/mL using PBS (phosphate buffer saline) buffer. The diluted suspension was spread on MRS agar, anaerobically incubated at 37 °C for 24 h, and the concentration of bacteria was determined. LGG were adjusted to 6×10^8 CFU/mL. The *A. senticosus* obtained from Changchun University of Chinese Medicine, was dried and particle through an 80-mesh sieve. Based on the preliminary experiments, the optimal fermentation conditions were as follows, the content of herbal medicine was 0.5 %. The inoculum of probiotics was 2 %, 4 %, 6 % and 8 %. The 0.5 % of herbal powder supplement with different concentrations of LGG were anaerobically incubated at 30 °C for 24 h. Mix different inoculum levels of ferment with the basal feed, and dry in an oven at 65 °C to prepare the fermented fish feed.

2.4. Experimental plan

The strategy of probiotic supplementation has been widely used in aquaculture. The fish were randomly divided into five groups. The fish treated with *A. senticosus* cultures fermented by LGG regarded as experimental group. The fish of control group was treated with basal diet without probiotic supplementation (con) and a 0.5 % *A. senticosus* control group (con-0.5 % As). The fish administration with basal diet coated by probiotic fermented cultures at 3 %–4 % of body weight twice a day at 09:00 a.m. and 16:00 p.m. for 8 weeks, including 2 % LGG +0.5 % As (LGG-2), 4 % LGG + 0.5 % As (LGG-4), 6 % LGG + 0.5 % As (LGG-6) and 8 % LGG + 0.5 % As (LGG-8).

2.5. Sample collection

Fish were randomly selected from each group at 0, 7, 14, 21, and 28 days post administration. Blood samples were collected from the caudal

Table 1
Primers used for qRT-PCR determination of differently cytokine genes.

Gene	Sequence (5′–3′)	Accession	PCR product (bp)
MyD88	F:GCCGACTCGCAAGTTGACTT R: GCTTTTTCCTCACGCTCATGT	GR679416.1	169
IL-10	F:AACTGATGACCCGAATGGAAAC R:CACCTTCTCCCAGTCGTCAAA	JX524550.1	143
IL-1β	F:AACTGATGACCCGAATGGAAC R:CACCTTCTCCCAGTCGTCAAA	AJ249137.1	133
IFN-γ	F: AACAGTCGGGTGTCGCAAG R:TCAGCAAACATACTCCCCAG	EU069818.1	141
TNF-α	F: TTATGTCGGTGCGGCCTTC R: AGGTCTTTCCGTTGTCGCTTT	EU909368.1	101
β-actin	F:AATGTGGTGCGTCTGTGCTT R:TGTTGTCATAGATGGGGTTGGA	AB039726.2	100

vein using a 1 mL syringe after anesthetized with 300 mg/L MS-222 (Sigma) for 15 min. The blood samples were incubated at 4 °C for 8 h, and the serum was obtained by centrifugation (2000×g, 10min, 4 °C) and stored at −80 °C. Furthermore, the pieces of spleen, liver, gills, kidney and intestine were rapidly dissected, frozen in liquid nitrogen, and stored at −80 °C, until RNA extraction.

2.6. Estimation of growth performance

The growth performance of fish treated with probiotic fermented cultures was monitored as described previously [25]. In brief, three fish were randomly sampled to measure the body weight and total length after administration for the evaluation of growth performance. Growth and nutrient efficiency were calculated following by the equations as mentioned below:

Weight gain rate (WGR, %) = (final weight-initial weight)/initial weight × 100.

Specific growth rate (SGR, %) = [(ln (final weight) - ln (initial weight)/time duration (days)] × 100.

2.7. Non-specific immune parameters

The serum samples collected at 0, 7, 14, 21and 28 day were used for determining the catalase (CAT), malondialdehyde (MDA), lysozyme (LZM), superoxide dismutase (SOD), glutathione peroxidase (GSH-PX), immunoglobulin M (IgM) and complement 3 (C3), complement 4 (C4) activity by using ELISA kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's instructions. All analyses were conducted in triplicates [26].

2.8. RNA isolation and real-time quantitative PCR

Total RNA was extracted from the liver, spleen, kidney, intestine and gills tissues using High Pure RNA Tissue Kit (BioFlux, China) following the manufacturer's instructions. The isolated RNA was reverse transcribed for cDNA synthesis using Reverse Transcriptase M-MLV kit (Takara, Japan). The qRT-PCR was conducted using the 7500 system (Applied Biosystem, USA) for evaluated the gene expression of TNF-α, IFN-γ, MYD88, IL-1β and Il-10. The β-actin gene was used as an internal control gene for qRT-PCR. The information on the GenBank registration of the target genes and the sequences of the primers are shown in Table 1. Each reaction was conducted in a final volume of 20 μL containing 10 μL SYBR Green Master Mix (Takara, Japan), 8 μL of sterile water, 0.5 μL of both forward and reverse primers, and 1 μL of cDNA as template. Each sample was repeated three times and quantified by the $2^{-\Delta\Delta Ct}$ method as described by Tattiyapong et al. [27].

2.9. Challenge

After 28 days of immunization, to evaluate the protective effect of the *A. senticosus* fermented cultures by probiotic, the crucian carp were intraperitoneally injected with pathogenic *A.hydrophila* TPS-30. The crucian carp fasted for 48 h before challenge experiment. The strain *A. hydrophila* TPS-30 ($LD_{50} = 1 \times 10^6$ CFU/mL) was incubated overnight and diluted to a concentration of 1×10^7 CFU/mL. Twenty fish were randomly selected from each group and intraperitoneally injected with 200 μL bacterial suspension. The survival rate of each group was determined after 14 days of monitoring [28]. Mortality was recorded and relative percentage survival (RPS, %) was calculated using the following formula. RPS= (1-% mortality in immunization group/% mortality in control group) × 100.

Table 2
Effects of *A. senticosus* cultures fermented by *L. rhamnosus* on weight gain rate (WGR) and specific growth rate (SGR) of crucian carp.

Groups	IBW(g)	FBW(g)	WGR (%)	SGR (%)
control	20.35 ± 0.32	24.51 ± 0.25	20.45 ± 0.52	14.86 ± 0.14
con-0.5% As	20.42 ± 0.62	30.63 ± 0.38	50.27 ± 1.32	36.4 ± 0.8
LGG-2	20.78 ± 0.32	29.49 ± 0.57	41.8 ± 0.6*	31.0 ± 0.9*
LGG-4	20.63 ± 0.21	32.52 ± 0.28	57.65 ± 0.25*	42.45 ± 0.25*
LGG-6	20.15 ± 0.52	37.56 ± 0.25	86.45 ± 3.55**	62.15 ± 0.95**
LGG-8	20.58 ± 0.47	35.63 ± 0.27	73.15 ± 2.65**	53.75 ± 0.75**

Note : IBW:initial weight. FBW : final weight. Weight gain rate (WGR, %) = (final weight-initial weight)/initial weight × 100. Specific growth rate (SGR, %) = [(ln (final weight) - ln (initial weight)/time duration (days)] × 100.

2.10. Histopathological assessment

The liver, spleen, kidney and intestine tissues of crucian carp were sampled after challenged, and fixed in 4 % neutral buffered formalin for 24 h and embedded in paraffin for histopathological analysis. Subsequently, the tissues were dehydrated in graded ethanol (70 %, 80 %, 85 %, 90 %, 95 % and 100 %) and washed with xylene as described using a tissue embedding machine (Leica Microsystems, Wetzlar, Germany) and prepared the serial sections (5 μm thick). Samples were stained with hematoxylin and eosin (H&E), for light microscopy analysis [29].

2.11. Statistical analysis

All statistical analyses were performed using SPSS 22.0 software and Graphical Prism v8.0. One-way analysis of variance (NOVA) followed by Tukey's test were used to analyze differences between different groups. Differences between experimental groups were considered as significant at ($p < 0.05$). Data are presented as the mean ± standard deviation (SD).

3. Result

3.1. Growth performance

The growth performance of fish treated with LGG fermented *A. senticosus* cultures are shown in (Table 2). After 28 days of administration, the WGR and SGR of the LGG-2 and LGG-4 groups were significant higher than the control group ($p < 0.05$). Also, the WGR and SGR of the LGG-6 and LGG-8 groups were significantly higher than the control group ($p < 0.01$), which have further enriched the variety of strains for probiotics fermentation of herbal medicine.

3.2. Humoral immune parameters

Humoral immune parameters of crucian carp treated with feeds containing LGG fermented *A. senticosus* cultures were determined. During the feeding period, serum CAT activity of fish in LGG-8 group

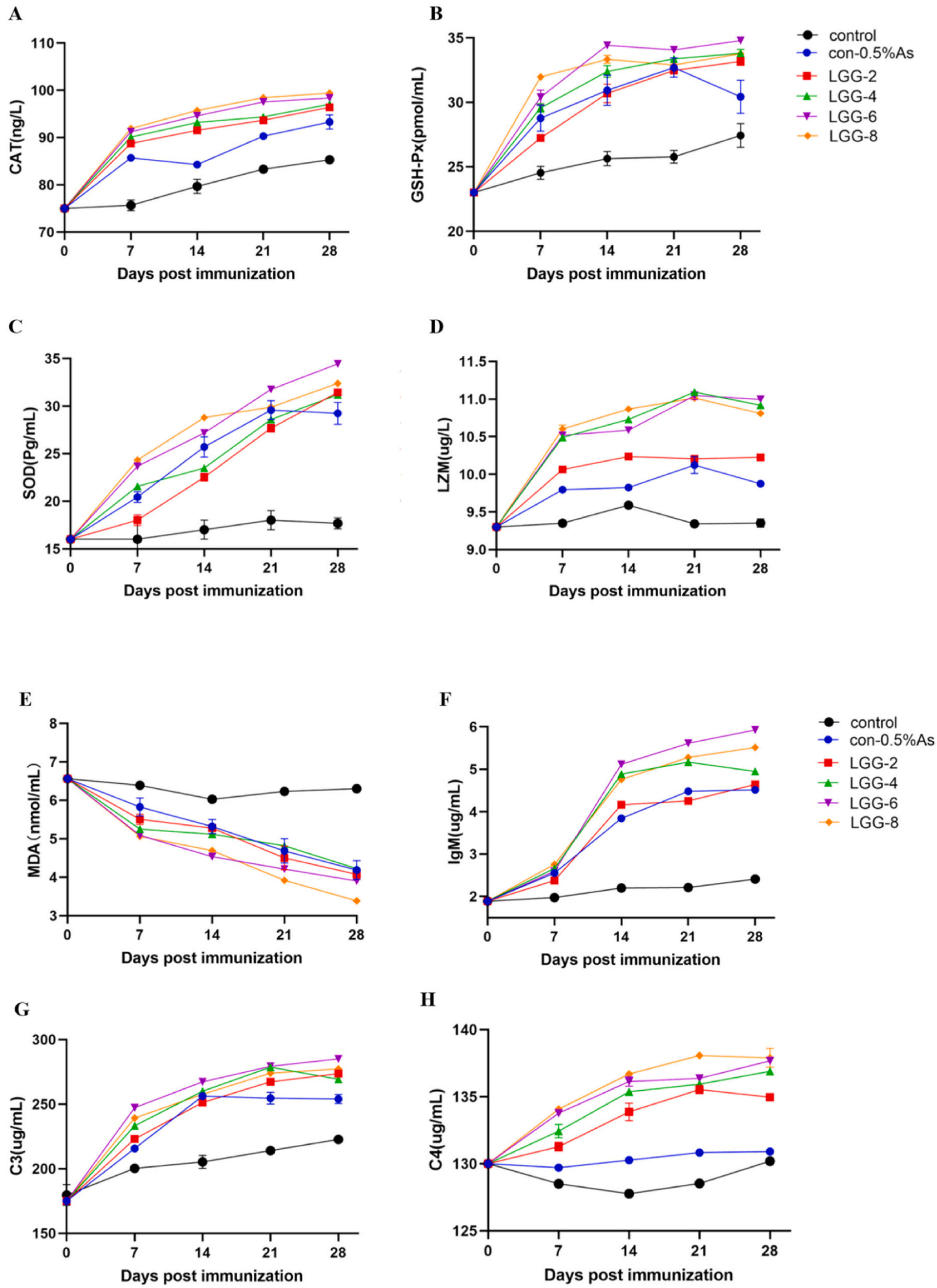

Fig. 1. Innate immune response parameters levels of CAT (A), GSH-Px (B), SOD (C), LZM (D), MDA (E), IgM (F), C3 (G) and C4 (H) stimulated by LGG fermented *A. senticosus* cultures in crucian carp (n = 3 fish/group).

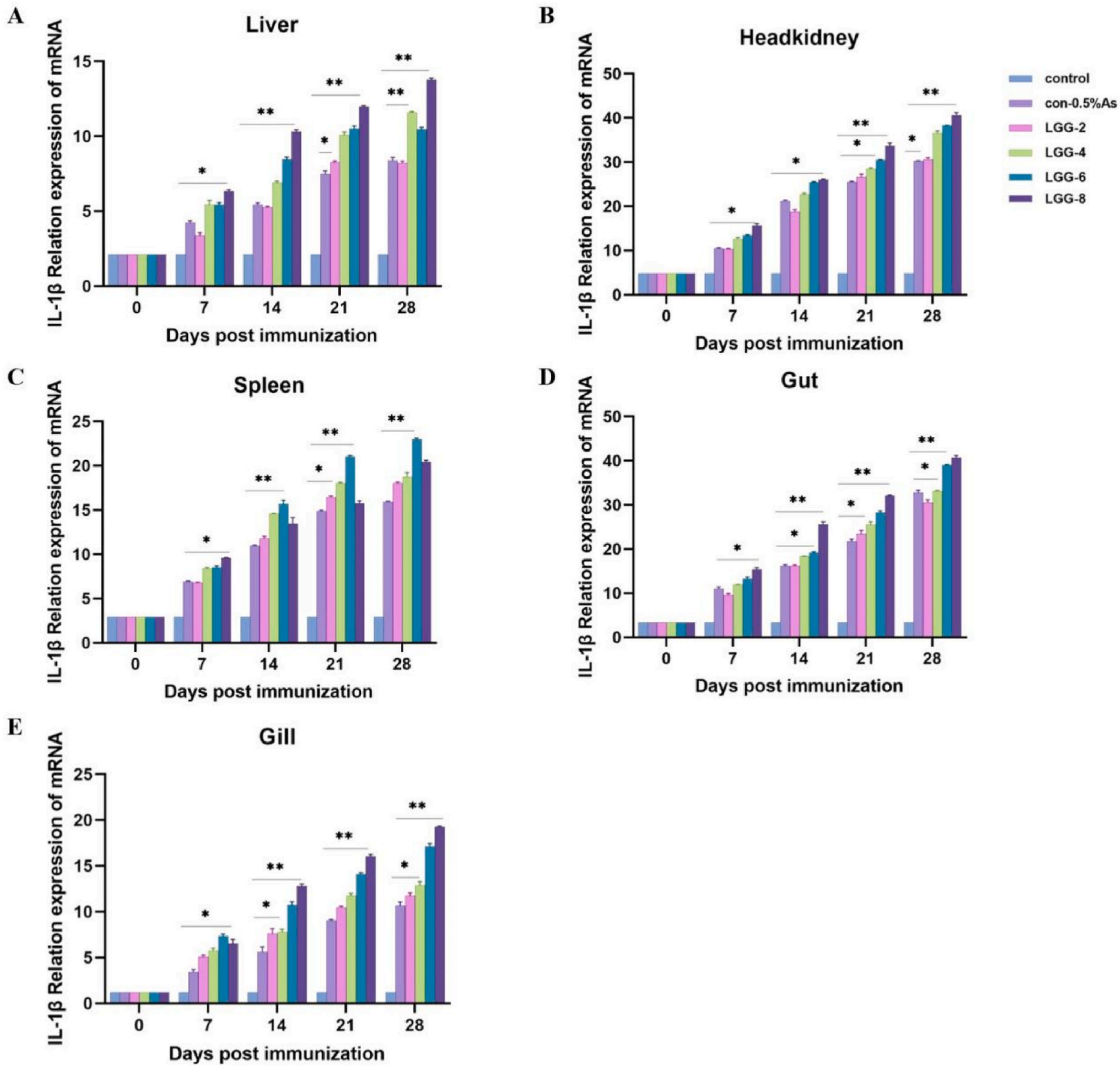

Fig. 2. qRT-PCR analysis of the expression of IL-1β gene in liver (A), headkidney (B), spleen (C), intestine (D), and gill (E) of crucian carp (n = 3 fish/group) received with LGG fermented *A. senticosus* cultures. The values are expressed as the means ± SD (n = 3).

was significantly increased compared with the control group at day 8 and day 10 (Fig. 1A). The fish serum CAT activity of LGG-2, LGG-4 and LGG-6 groups was up-regulated during the trial (Fig. 1A). Similarly, higher concentration of serum GSH-PX was observed at day 28 in probiotic fermented cultures groups compared with the control group (Fig. 1B), with a significant difference at day 28 ($p < 0.01$). The results showed that SOD activity in fish received with LGG fermented cultures was closely associated with probiotic proportion in fermented product (Fig. 1C). We found that serum SOD activity of fish in LGG-2 and LGG-4 groups lower than the con-0.5 % As group at 21 day, but higher than the control group at 28 day ($p < 0.05$) (Fig. 1C). The fish serum SOD activity of LGG-6 group reached peak on day 28 after treatment, which was significantly higher than the control group ($p < 0.01$) (Fig. 1C). Moreover, our data revealed that LGG fermented *A. senticosus* cultures could improve the level of serum LZM in fish, and the serum LZM concentration of LGG-4, LGG-6 and LGG-8 groups fish was significant higher than the control group at day 21 ($p < 0.01$) (Fig. 1D).

As shown in Fig. 1E, the MDA activity of all groups treated with probiotic fermented product showed a down-regulated trend, which was significant different compared to the control group ($p < 0.05$). Further, significant difference of MDA activity was observed between LGG-8 and control groups ($p < 0.01$) (Fig. 1E). In contrast, the serum IgM level of LGG-2 group was significant higher than the control group at day 28 ($p < 0.05$) (Fig. 1F). At 28 days of immunization (Fig. 1F), the IgM level of LGG-2 group was significantly different from the control group ($p < 0.05$). The IgM level of the LGG-4 group was initially elevated and then decreased during the trial with no significant difference from the control group ($p < 0.05$) (Fig. 1F). IgM levels in the LGG-6 and LGG-8 groups were stable and significantly differences with the control group after 14 days ($p < 0.01$).

Besides, the serum C3 level of fish treated with probiotic fermented product stably increased, however, the concentration of C3 down-regulated after 14 days treatment (Fig. 1G). The C3 level of LGG-6 group reached peak on the 28th day, and dramatic differences was found in the C3 concentration of LGG-6 and control groups ($p < 0.01$). As shown in (Fig. 1H), there was a clear trend of increasing serum C4 level in probiotic fermented cultures groups. From Fig. 1H, it could be seen that C4 level of LGG-4 and LGG-6 groups decreased on day 14, however, there was a significant up-regulation of LGG-4 and LGG-6 groups as compared to control group ($p < 0.01$). Further analysis revealed that the C4 level of LGG-2 and LGG-8 groups significant higher than the control group at day 21 ($p < 0.001$).

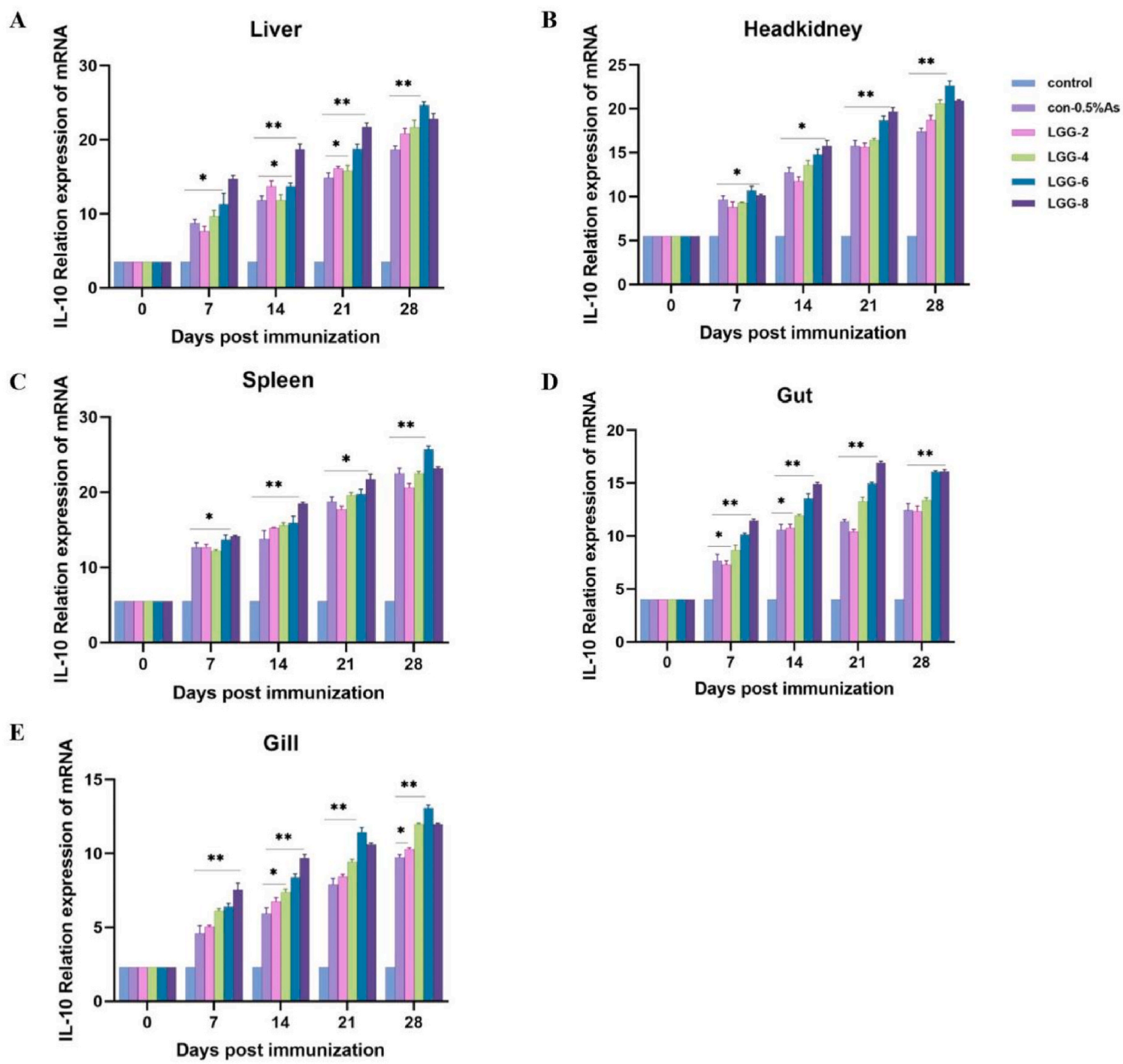

Fig. 3. qRT-PCR analysis of the expression of IL-10 gene in liver (A), headkidney (B), spleen (C), gut (D), and gill (E) of crucian carp (n = 3 fish/group) received with LGG fermented *A. senticosus* cultures. The values are expressed as the means ± SD (n = 3).

3.3. Immune gene expression

To investigate the mRNA expression levels of immune-related genes in crucian carp liver, head-kidney, spleen, intestine, and gill tissues administration with LGG fermented cultures, we examined five immune-related genes expression by qRT-PCR. As shown in Fig. 2A, the 2 %, 4 %, 6 %, and 8 % probiotic fermented product induced consistent expression patterns of the IL-1β gene. Our data suggested that the mRNA level of IL-1β was up-regulated until 28 day in the liver and head-kidney, which was significantly higher than the control group ($p < 0.01$) (Fig. 2A and B). The expression of IL-1β gene in the spleen observed in LGG-6 group showed an obvious enhancement compared with the control group during the trial ($p < 0.01$) (Fig. 2C). In the liver, spleen, intestine, and gill, a significant IL-1β gene expression increased of LGG-8 group was detected at day 28 compared with the control group ($p < 0.01$) (Fig. 2A–C, D, E), and IL-1β transcription of LGG-6 and LGG-8 groups reached a peak in head-kidney and intestine tissues at day 28.

The increased IL-10 transcripts were obtained in fish fed diets containing LGG fermented cultures throughout the 28 days study period (Fig. 3). A higher IL-10 expression in all tissues was observed after 7 days feeding ($p < 0.05$) of LGG-2, LGG-4, LGG-6, and LGG-8 groups compared to the control group (Fig. .3A–E). Besides, whereas a slight changes of IL-10 expression was detected in the various tissues after 21 days of immunization (Fig. 3).

The highest expression levels of IFN-γ gene in the spleen was observed at day 28 post immunization from LGG-6, and LGG-8 treatments ($p < 0.01$) (Fig. 4C). After 28 days of oral administration, the IFN-γ gene expression in the liver, spleen and gills of LGG-6 and LGG-8 groups was significantly up-regulated as compared to the control group ($p < 0.05$) (Fig. 4A–C, E). An obvious enhancement of IFN-γ expression in the mucosal-associated tissues, including the intestine and gills of LGG-6 group was found, which was higher than LGG-8 group (Fig. 4D and E). Moreover, the mRNA level of IFN-γ gene in the liver, head-kidney, intestine and gills tissues of LGG-2 group were lower than the control group at sampling time (Fig. 4A, B, D, E).

As shown in Fig. 5, the expression of the TNF-α gene in various tissues was up-regulated through LGG fermented cultures treatment. The mRNA level of TNF-α gene in the LGG-2 and LGG-4 groups showed an increasing trend in detected tissues, but the increasing trend was not significant ($p < 0.05$) (Fig. 5A–E). In contrast, the transcription level of TNF-α gene in all tissues of LGG-6 group was significantly higher than the control group ($p < 0.01$) (Fig. 5A–E). Interestingly, a higher TNF-α expression was observed after 28 days of treatment in liver, head-kidney, spleen and gills compared to the LGG-8 group ($p < 0.05$)

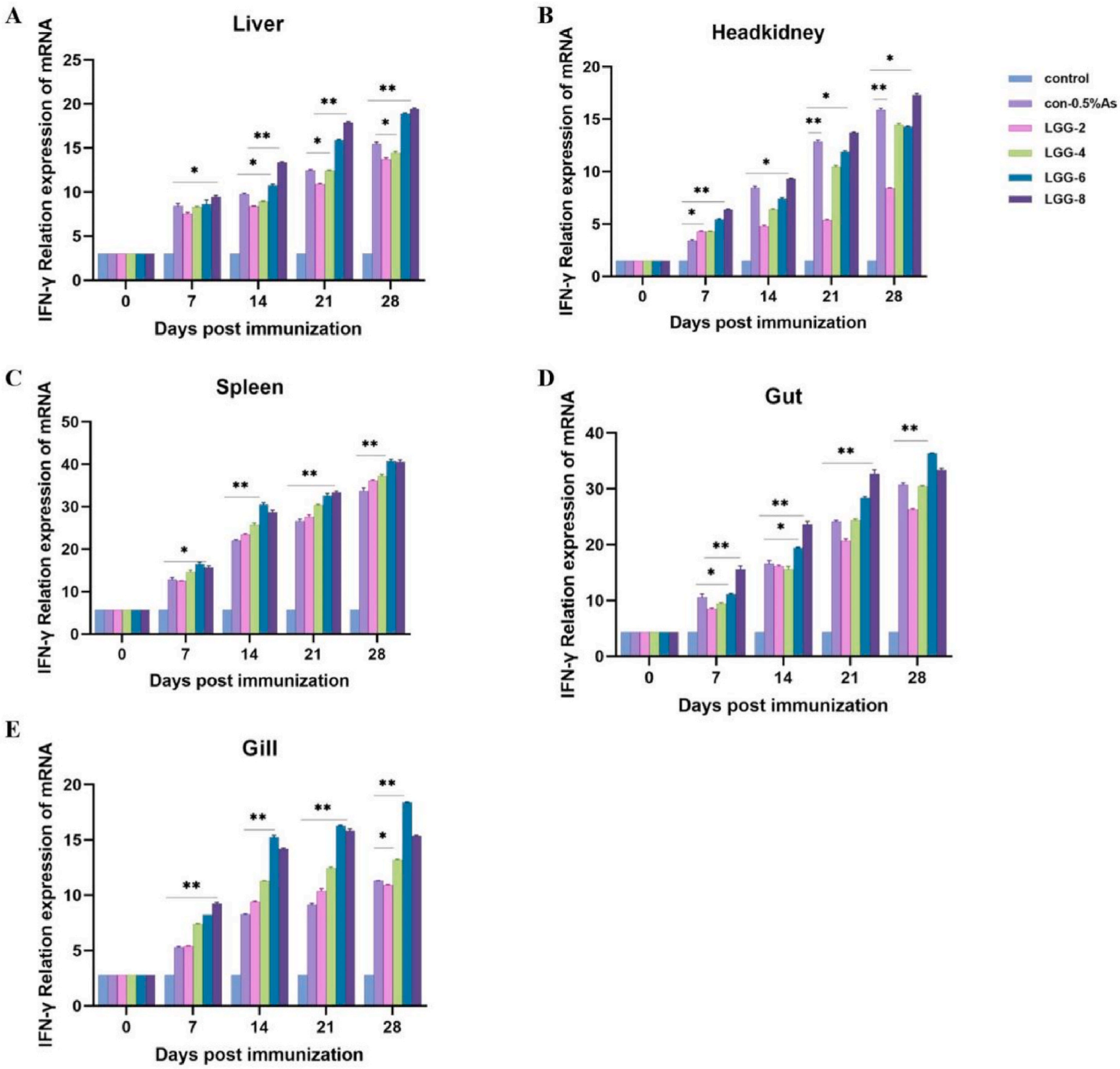

Fig. 4. qRT-PCR analysis of the expression of IFN-γ gene in liver (A), headkidney (B), spleen (C), gut (D), and gill (E) of crucian carp (n = 3 fish/group) received with LGG fermented *A. senticosus* cultures. The values are expressed as the means ± SD (n = 3).

(Fig. 5A-C, E). Besides, compared with control, the mRNA expression of TNF-α gene was significantly higher in LGG-8 group after 4 weeks of feeding in head-kidney, spleen and intestine tissues ($p < 0.01$) (Fig. 5B–D).

Our results revealed that the mRNA level of cytokine MyD88 was up-regulated until 28 day after fish received with probiotic fermented cultures (Fig. 6). And the fermented product (LGG-2 and LGG-4 groups) enhanced the expression of MyD88 gene in different tissues after 4 weeks treatment compared to the control group ($p < 0.05$) (Fig. 6A–E). Similarly, the increased MyD88 transcripts were found in fish treated with diets containing 6 % probiotic fermented cultures ($p < 0.01$) (Fig. 6B–D, E).

3.4. Histological analysis

Histological examination of fish fed the probiotic fermented diet for 4 weeks showed a reduction in liver, spleen, head-kidney, and intestinal inflammatory damage after challenged with *A. hydrophila* in comparison with the control group (Fig. 7). As shown in Fig. 7S-X, the inflammatory damage of fish treated with fermented cultures was improved after *A. hydrophila* infection compared to the control group. The complete intestinal villi and epithelial cells were observed in the probiotic fermentation groups, with slight haemorrhage (Fig. 7S–X). Conversely, the intestinal villi of crucian carp in the control group were disordered, which exacerbated the development of the epithelial cells shed and muscle layer structure loose.

In Fig. 7A-F, our data revealed that the arrangement of liver cells in the control group was loose and irregular, and there was evident inflammatory infiltration of cells in the hepatic lobules. Similarly, there was significant haemorrhage and evident lobectomy margins in the LGG-2 and LGG-4 groups (Fig. 7C and D). Additionally, less significant haemorrhage was found in the LGG-6 and LGG-8 groups (Fig. 7E and F), suggesting that the hepatocytes are round or ovoid, densely packed, with large and round nuclei.

More hyperaemia was found in the spleen tissue of the control group fish after 72 h post infection (Fig. 7G). Similar with this result, the head-kidney structure of the control group fish was damaged, which also disturbed the arrangement of the renal tubules (Fig. 7M). Besides, the bacterial infection leaded to the number of glomeruli and renal tubules decreased and tissue haemorrhage. Reciprocally, the glomeruli and renal tubules were clearly distinguishable, and the arrangement was orderly and clear in the head-kidney of crucian carp received with probiotic fermented cultures (Fig. 7O–R).

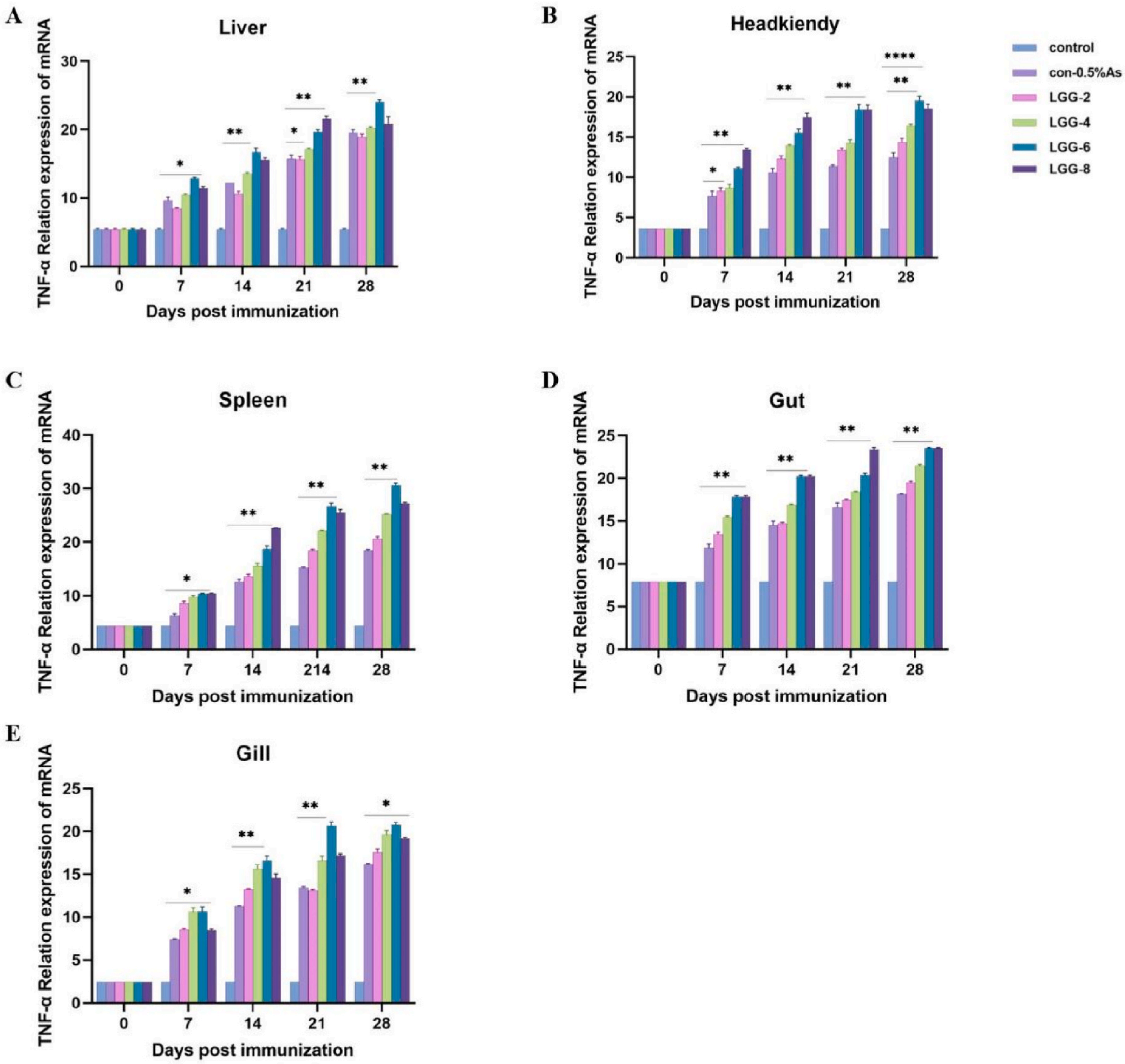

Fig. 5. qRT-PCR analysis of the expression of TNF-α gene in liver (A), headkidney (B), spleen (C), gut (D), and gill (E) of crucian carp (n = 3 fish/group) received with LGG fermented *A. senticosus* cultures. The values are expressed as the means ± SD (n = 3).

3.5. Evaluation of protective efficacy

Two weeks post *A. hydrophila* infection, the protective effect of the probiotic fermented cultures on crucian carp against bacterial infection was determined as shown in Fig. 8. During infection, there was a significant increase in mortality in the control group (100 %). In addition, the con-0.5 % As group reached a basic stability on the 8th day after challenged, and the survival rate was 45 %. Further analysis suggested that the protection rate of the LGG-2 group (40 %) was lower than the con-0.5 % As group, while the other probiotic fermentation groups was higher than the control group (50 %, 60 %, and 50 %). All crucian carp showed signs of haemorrhaging, mainly in the gills, thorax, abdomen and caudal fin base, and appeared to die (data not shown).

4. Discussion

Aeromonas species is widely distributed in aquaculture water, with a broad spectrum of pathogenicity in freshwater fish [30]. *A. hydrophila* is a Gram-negative, concurrent anaerobic, motile, mesophilic, oxidase-positive bacillus, which is found worldwide in aquatic environments. Several virulence factors expressed by *A. hydrophila*, such as proteases1, and which could be the aetiological agent of gastrointestinal infections, wound infections, infections of the biliary system, post-traumatic cellulitis and bacteraemia [31]. It is vital to develop novel, residue-free and non-toxic products for the prevention and treatment of diseases caused by *A. hydrophila* in aquaculture under antibiotic restrain background. In recent years, vaccine application has attracted a great deal of attention due to its non-toxicity and high level of safety [32].

The inactivated vaccines developed based on three bacterial pathogens in Japanese eels (*Anguilla japonica*), whereas, which are not stable in water environment and easily lead to the diffusion of pathogens [33, 34]. Therefore, the probiotic with high biosecurity for the prevention of pathogenic bacteria in aquaculture has been considered as potential strategy. Previous studies have shown that the engineered *Lactobacillus casei* as an oral vehicle could improve the survival rate of common carp against *A. veronii* [28]. Four *Bacillus velezensis* strains isolated from *C. auratus* with broad-spectrum antibacterial activity against various pathogens reported by Zhang et al., demonstrating that *Bacillus* spp. Could enhance local mucosal immune response through resistance against pathogen colonization [35]. However, the application of probiotic alone for the treatment of bacterial infection could lead to a loss of microbial metabolites. The majority of probiotics are lost in the gastrointestinal tract, and the reduced colonization affects their role in

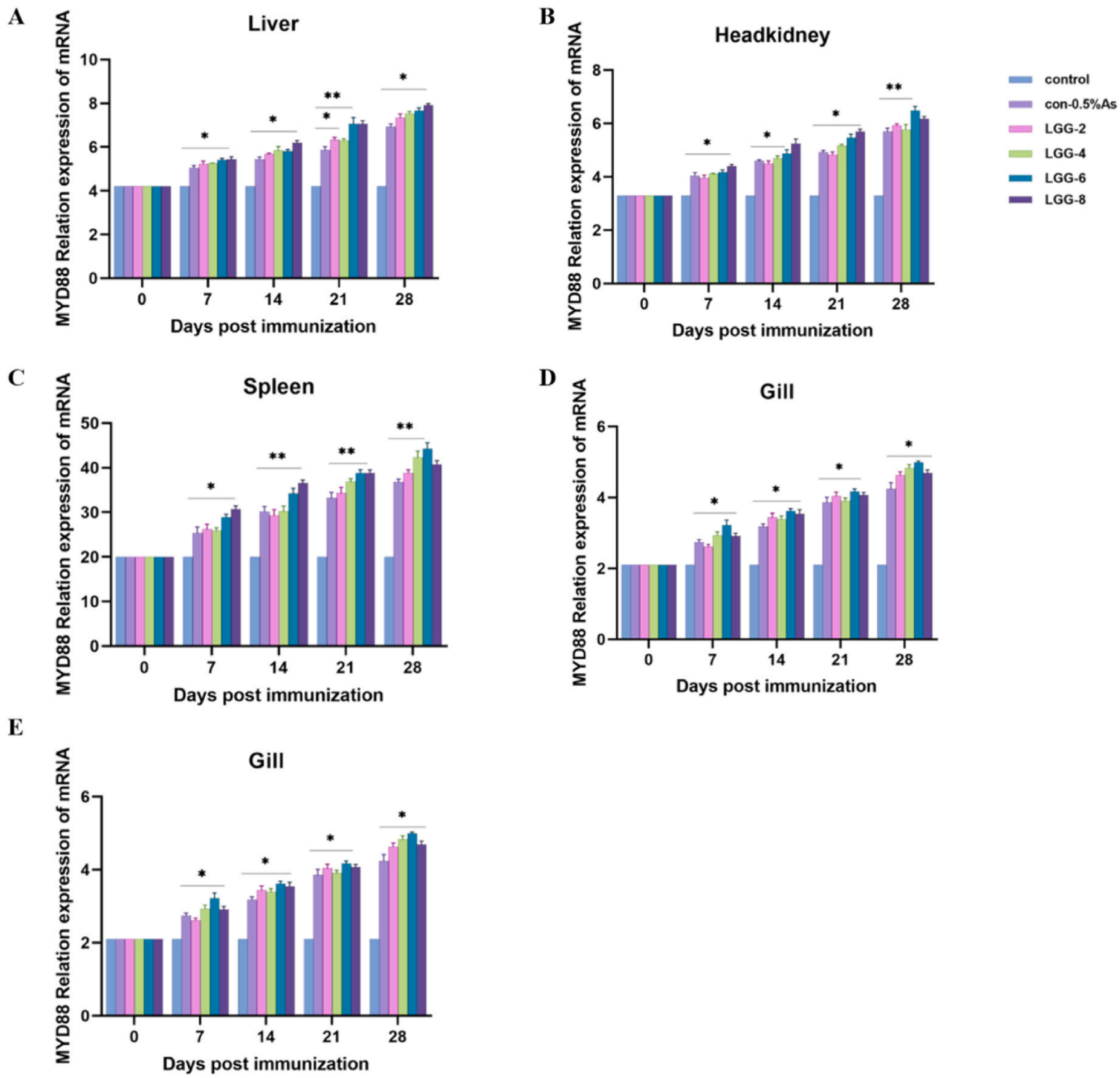

Fig. 6. qRT-PCR analysis of the expression of MyD88 gene in liver (A), headkidney (B), spleen (C), gut (D), and gill (E) of crucian carp (n = 3 fish/group) received with LGG fermented *A. senticosus* cultures. The values are expressed as the means ± SD (n = 3).

treatment [36].

Fermented traditional Chinese medicine can utilize the acid-producing probiotic, which can be contributed to increase active ingredients and alleviate toxic or side effects. The application of fermentation containing traditional Chinese medicine and probiotic is considered as a novel anti-bacterial drug [37]. Wang et al. found that Chinese herbal medicine *Sanguisorba officinalis* fermented by three probiotics could protect fish against *A. hydrophila* infection, which also improved the cytokines expression in mucosal-associated tissues [38]. Likewise, fermented *Astragalus* cultures diets could improve growth performance by regulate the intestinal microbiota in chicken model [39]. Similar with our work, the weight gain and specific growth rate of crucian carp could be up-regulated through receiving with cultures of *L. rhamnosus* fermented with *A. senticosus* (Table 2). Moreover, Gao et al. prepared a novel probiotic formulation, namely Chinese medicine-probiotic compound microecological preparation (CPCMP), which is composed of five traditional Chinese medicine herbs (*Galla Chinensis*, *Andrographis paniculata*, *Arctii Fructus*, *Glycyrrhizae Radix*, and *Schizonepeta tenuifolia*) fermented by *Aspergillus niger*, *L. plantarum* A37 and *L. plantarum* MIII. The CPCMP significantly increased the average body weight and average daily gain as compared to the control group ($p < 0.05$), and further decreased the ratio of feed and gain in broilers [40]. In mice, oral probiotic fermented red ginseng has been reported to reduce the severity of colitis. Notably, there was no obvious pathological damage in the colon tissue observed in the mice that antibiotic-associated diarrhea treated with fermented red ginseng. In this work, more complete crypt structures and less damage of intestinal villi in *L. rhamnosus* fermented cultures treated fish were found after *A. hydrophila* infection, which indicated that the traditional Chinese medicine fermented by probiotic could promote active ingredients release and further improve intestinal digestion and absorption function (Fig. 7) [41].

The non-specific immune-related molecules can protect fish from pathogen in complex water environment, such as IgM, complement C3, and C4 [42]. Our data reveal that the levels of C3 and C4 in the serum of fish administrated with *L. rhamnosus* fermented cultures significantly increased, which implied that innate immune response could be triggered by probiotic fermented product [43]. In freshwater fish, oxidative stress damage can be caused during bacterial infection [44]. Thus, we further evaluated the antioxidant ability of crucian carp after *A. senticosus* fermented product treatment. We found that higher level of the CAT, GSH-PX, SOD activity in fish treated with *L. rhamnosus*

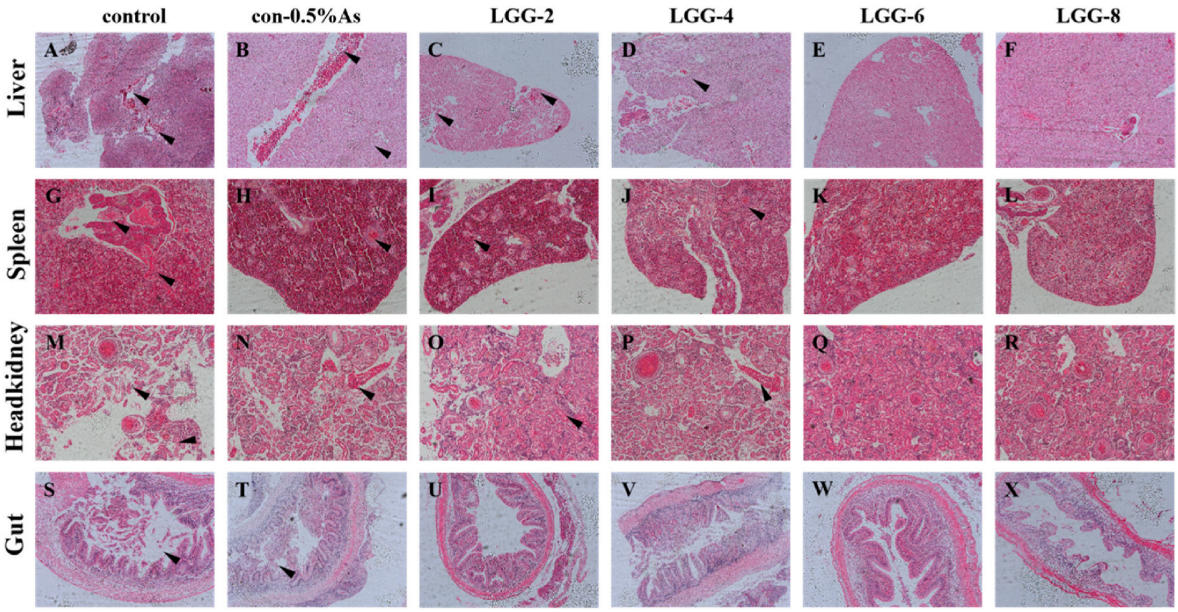

Fig. 7. Histopathological sections of liver (A–F), spleen (G–L), headkidney (M–R) and gut (*S*–X) of crucian carp fed fermented herbal LGG cultures after pathogen infection. The black arrow indicates the site of hemorrhagic lesions in crucian carp after challenged with *A. hydrophila*. Data are representative of three independent experiments (Scale bars, 50 μm).

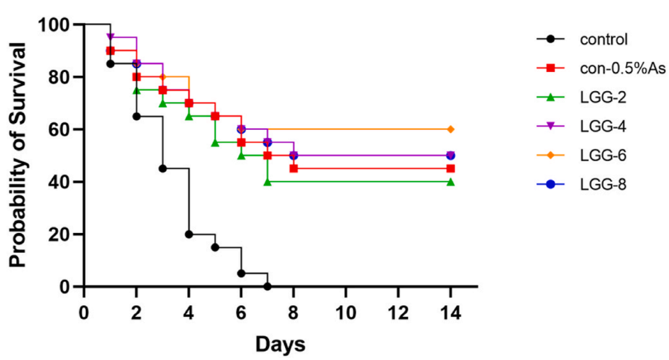

Fig. 8. Survival rate of fish oral administration with *L. rhamnosus* fermented *A. senticosus* cultures and PBS following challenged with the *A. hydrophila* strain. Twenty fish per group were used to record the survival rate for 14 days.

fermented cultures, which explored that probiotic fermented cultures could scavenge oxygen free radicals through the enhancement of antioxidant enzyme activity [45]. Moreover, the serum MDA activity, as an indicator of lipid peroxidation, was significantly decreased in LGG-8 group fish, and we speculated that the acid-producing probiotic might promote the production of antioxidant stress substances. These results were accordance with previous study reporting that *Piper sarmentosum* extract, a kind of traditional Chinese medicine, could increase the serum GSH-PX activity and decrease MDA level in weaned piglets [46]. A similar pattern of results was obtained in lysozyme activity, the fish orally administration with different doses of *L. rhamnosus* fermented cultures induced higher lysozyme level in serum [47]. The present study confirmed the findings about innate immunity could be enhanced by probiotic fermented traditional Chinese medicine cultures.

Inflammatory cytokines generally regulate the activation, differentiation and homing of immune-related cells, ultimately controlling and eradicating intracellular pathogens, including bacteria, viruses and parasite [48]. In general, microbial antigens can induce IL-1β expression through Toll-like receptors (TLRs) and NOD-like receptors (NLRs). Of note, IL-1β first binds to IL-1R1 on the surface of target cells, after ligand binding, MyD88 interacts with IL-1R1 via TIR domain [49]. We described the results of cytokines transcription, which showed that the expressions of IL-1β and MyD88 genes up-regulated in various tissues of crucian carp (Figs. 2 and 6). The results of the present study found clear support for the higher expression of TNF-α gene in the spleen tissue of LGG-6 and LGG-8 groups (Fig. 5), which indicated that the pectinase, amylase produced by fermented probiotic might involve in the inflammation. Alternatively, it could simply mean that lactic acid bacteria accelerate the lysis of herbs and promote the release of *A. senticosus* polysaccharides, which might be contributed to regulate the development of inflammation [50]. This finding is consistent with what has been found in previous study, triterpene saponins extracted from *Acanthopanax* could improve anti-inflammatory response through inhibiting the activation of NF-κB signaling pathway by blocking the TLR4 receptor [51,52]. The present study demonstrated the findings about the mRNA transcription of IL-10 gene were up-regulated in fish received with high dose of *L. rhamnosus* fermented cultures (Fig. 3). The findings are directly in line with previous work that *L. rhamnosus* GG alleviated DSS-induced intestinal inflammation [53]. IFN-γ is produced by activated T cells and natural killer cells, which play a vital role in innate and adaptive immunity against the pathogen invasion [54,55]. It has been proven that fermented lactic acid bacteria can significantly enhance the anti-tumor ability through up-regulating the IFN-γ level [56]. These basic findings are consistent with research showing that probiotic fermented traditional Chinese medicine cultures could stimulate and improve the immune-related function in fish (Fig. 4).

More investigations confirmed that *A. hydrophila* infection can cause gastroenteritis, histopathological damage, and high mortality in freshwater fish [57,58]. Our finding showed that crucian carp infected with *A. hydrophila* can be prevented by feeding *L. rhamnosus* fermented cultures at different doses. Notably, the survival rate of fish treated with probiotic fermented production was higher than control group (Fig. 8). In line with previous research, Noshair et al. demonstrated that adding probiotics to the feed can enhance the resistance of Nile fish to the infection of *A. hydrophila* and improve the survival rate of fish [59]. Furthermore, severe haemorrhage was observed in the liver, spleen and headkidney tissues of the control and low-dose treatment groups (Fig. 8). These results supported the view of Ifeanyi et al. that diseased broiler chickens fed diets supplemented with *Bacillus* showed an improvement in the impairment of intestinal villi compared to the control group [60].

In conclusion, we explored the protection of *A. senticosus* fermented cultures by *L. rhamnosus* against *A. hydrophila*, focusing on the immune-related parameters after long-term feeding of fermented products. We found that the *A. senticosus* fermented by high-dose *L. rhamnosus* can increase the activity of enzymes in serum, and improve anti-oxidation function of fish. Importantly, *A. senticosus* fermented cultures could provide stronger protection for fish against *A. hydrophila.* However, there are still few studies on the mechanism of probiotic fermented traditional Chinese medicine against pathogenic bacteria in fish. Thus, further researches are needed to evaluate the synergistic effect of herbal medicine and probiotic to regulate the immune response in fish. Future research should focus on the changes in the components of Chinese herbal medicines fermented by probiotic.

Author contribution

Y-HM, Y-DS, DZ are the primary investigators in this study. J-TL,X-FL, NL, YT, and HL participated in the animal experiments. Y-HM, PS, SAS, W-WS, LZ, X-FS, C-FW, A-DQ, and D-XZ designed this study and wrote the article. All authors contributed to the article and approved the submitted version.

Ethics statement

The animal study was reviewed and approved by the Jilin Agriculture University Institutional Animal Care and Use Committee (JLAU08201409) and carried out in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publications No. 8023).

CRediT authorship contribution statement

Yi-Han Ma: Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing. **Dong-Xing Zhang:** Data curation, Formal analysis, Investigation.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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